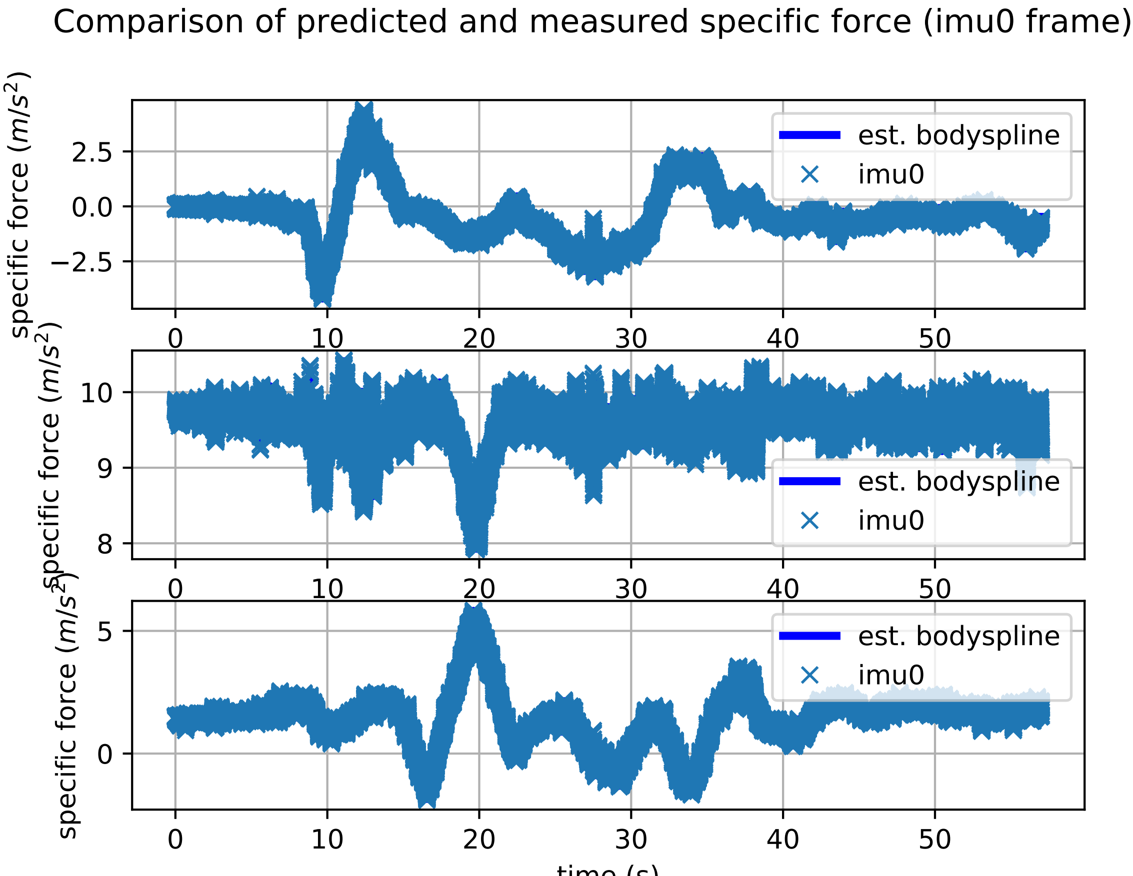
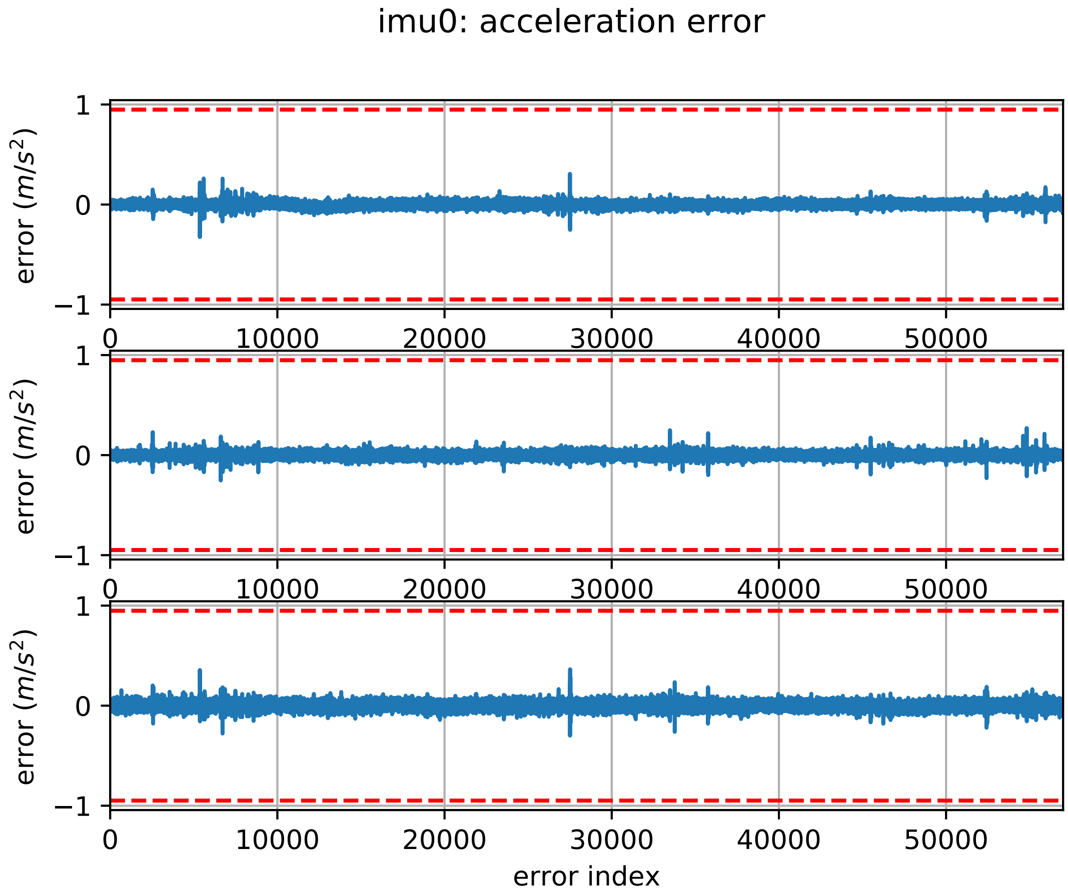
```
Calibration results
===================
Normalized Residuals
----------------------------
Reprojection error (cam0):     mean 0.174426975562, median 0.16442485528, std: 0.0904326802405
Reprojection error (cam1):     mean 0.161142017549, median 0.150899376395, std: 0.0854402180726
Gyroscope error (imu0):        mean 0.0830384621029, median 0.0754639187264, std: 0.0468897332687
Accelerometer error (imu0):    mean 0.132546551002, median 0.121931359413, std: 0.07146413775

Residuals
----------------------------
Reprojection error (cam0) [px]:     mean 0.174426975562, median 0.16442485528, std: 0.0904326802405
Reprojection error (cam1) [px]:     mean 0.161142017549, median 0.150899376395, std: 0.0854402180726
Gyroscope error (imu0) [rad/s]:     mean 0.00262590673643, median 0.00238637864337, std: 0.0014827835
Accelerometer error (imu0) [m/s^2]: mean 0.0419148997167, median 0.0385580813946, std: 0.022598944

Transformation (cam0):
-----------------------
T_ci:  (imu0 to cam0): 
[[-0.99527179 -0.09709206 -0.0026817   0.04714657]
 [-0.01018136  0.1317452  -0.99123133 -0.00296758]
 [ 0.09659399 -0.98651727 -0.13211081 -0.02053869]
 [ 0.          0.          0.          1.        ]]

T_ic:  (cam0 to imu0): 
[[-0.99527179 -0.01018136  0.09659399  0.04887735]
 [-0.09709206  0.1317452  -0.98651727 -0.01529325]
 [-0.0026817  -0.99123133 -0.13211081 -0.00552851]
 [ 0.          0.          0.          1.        ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.0281093978436


Transformation (cam1):
-----------------------
T_ci:  (imu0 to cam1):
```

```
 [ 0.          0.          0.          1.        ]]

T_ic:  (cam1 to imu0):
[[-0.99710869  0.00855313 -0.07550566 -0.08750697]
 [ 0.07597264  0.13254629 -0.98826092 -0.01560775]
 [ 0.00155527 -0.99113991 -0.13281286 -0.00589348]
 [ 0.          0.          0.          1.        ]]

timeshift cam1 to imu0: [s] (t_imu = t_cam + shift)
-0.0281142648647

Baselines:
----------
Baseline (cam0 to cam1):
[[ 0.98501364  0.01861932 -0.1714685  -0.13596553]
 [-0.01872394  0.99982418  0.00100727  0.00084646]
 [ 0.17145711  0.00221839  0.98518909 -0.01065707]
 [ 0.          0.          0.          1.        ]]
baseline norm:  0.136385166838 [m]



Gravity vector in target coords: [m/s^2]
[ 0.0046518   0.05832876 -9.80637543]


Calibration configuration
=========================

cam0
-----
  Camera model: pinhole
  Focal length: [238.88264329583697, 238.43988669054954]
  Principal point: [239.65578750366714, 321.1613847182054]
  Distortion model: equidistant
  Distortion coefficients: [0.004136165249475896, 0.09509478306213548, -0.11766502639762973, 0.0462
  Type: aprilgrid
  Tags:
```

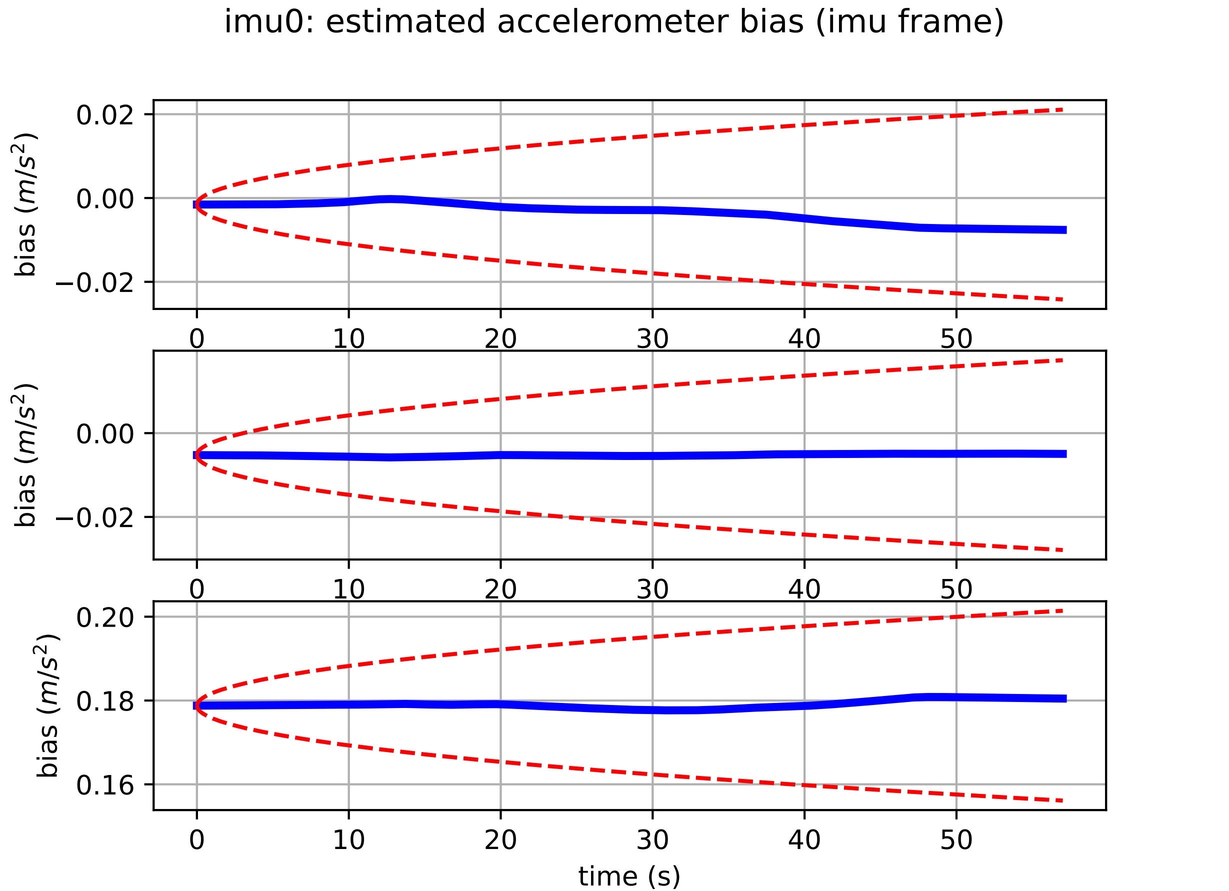
```
    Spacing 0.010372 [m]



cam1
-----
  Camera model: pinhole
  Focal length: [238.01314574429594, 237.9290215251219]
  Principal point: [238.5959141485548, 318.63275203667945]
  Distortion model: equidistant
  Distortion coefficients: [0.011015556563154302, 0.06646119044294033, -0.07637866453799633, 0.02562
  Type: aprilgrid
  Tags:
    Rows: 7
    Cols: 5
    Size: 0.02593 [m]
    Spacing 0.010372 [m]




IMU configuration
=================

IMU0:
----------------------------
  Model: calibrated
  Update rate: 1000.0
  Accelerometer:
    Noise density: 0.01
    Noise density (discrete): 0.316227766017
    Random walk: 0.001
  Gyroscope:
    Noise density: 0.001
    Noise density (discrete): 0.0316227766017
    Random walk: 0.0001
  T_i_b
    [[ 1.  0.  0.  0.]
     [ 0.  1.  0.  0.]
```

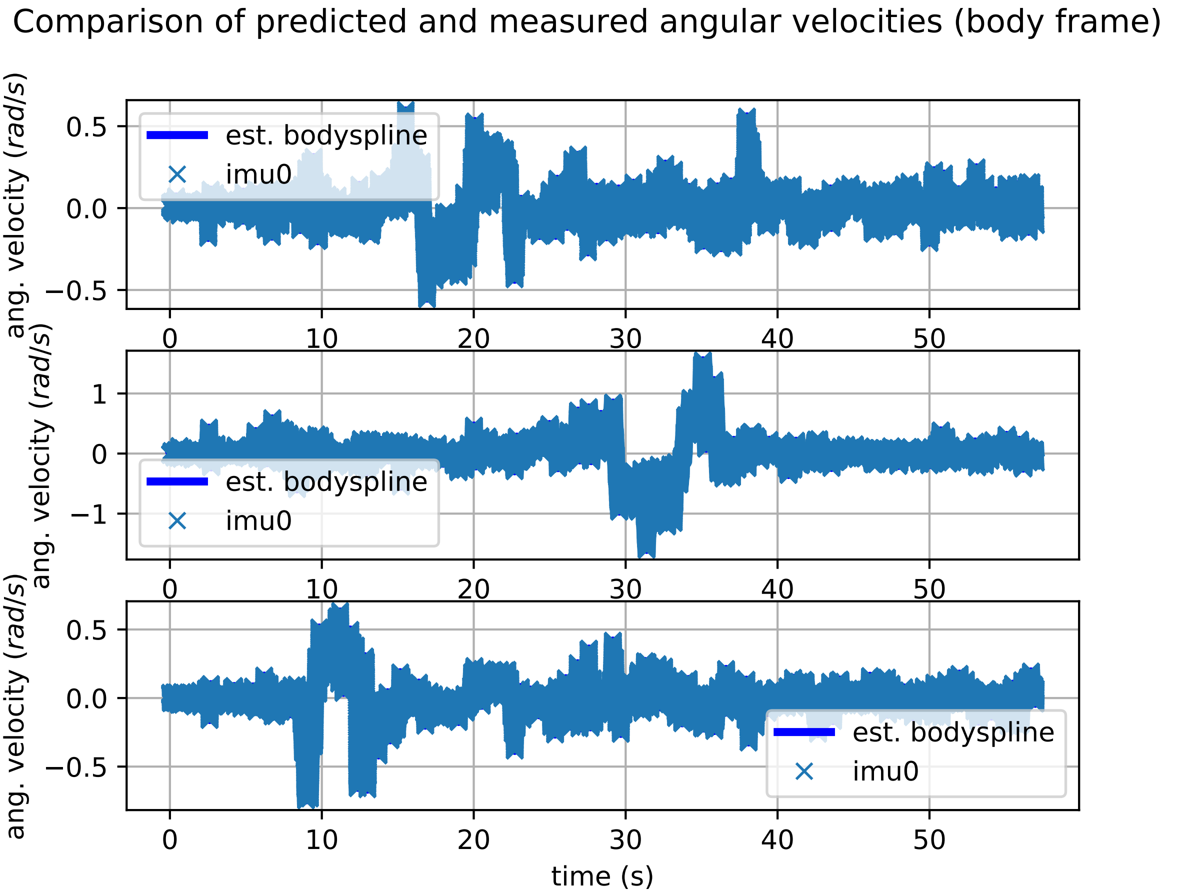


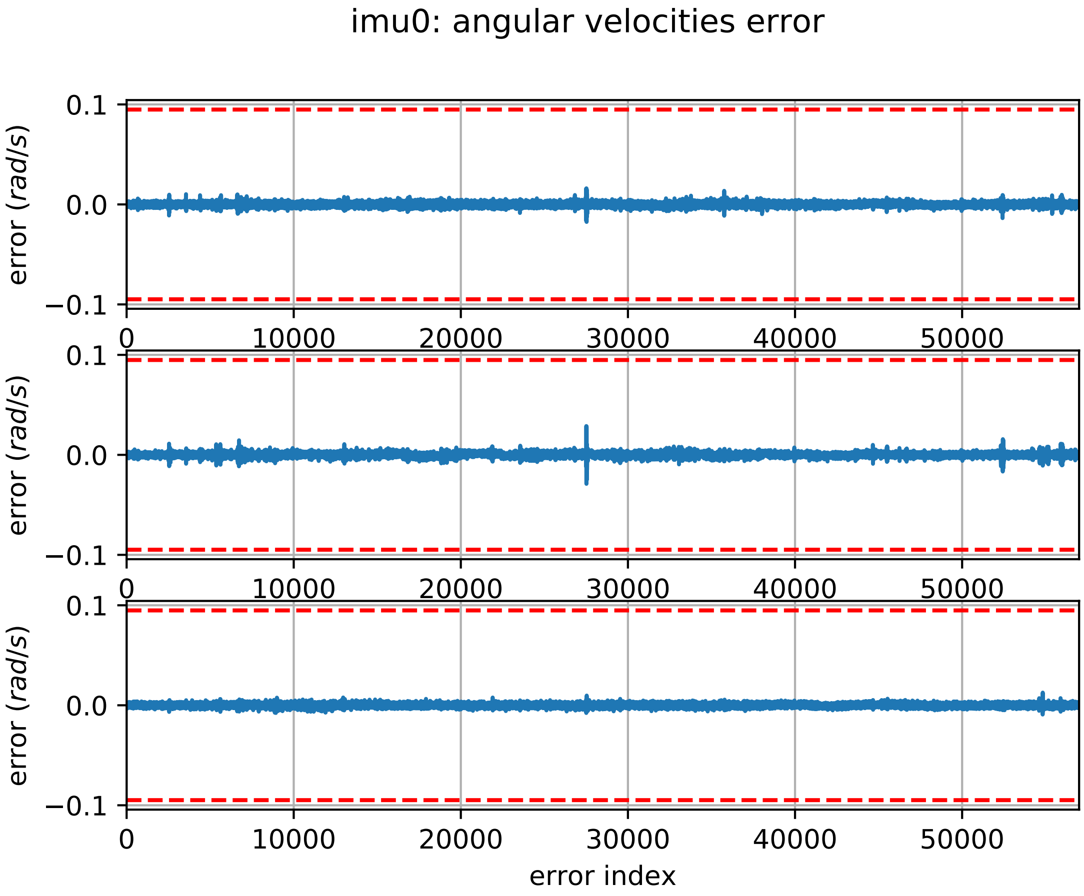
Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
est. bodyspline
imu0
2.5
0.0
−2.5
10
9
8
5
0
0
10
20
30
40
50
time (s)

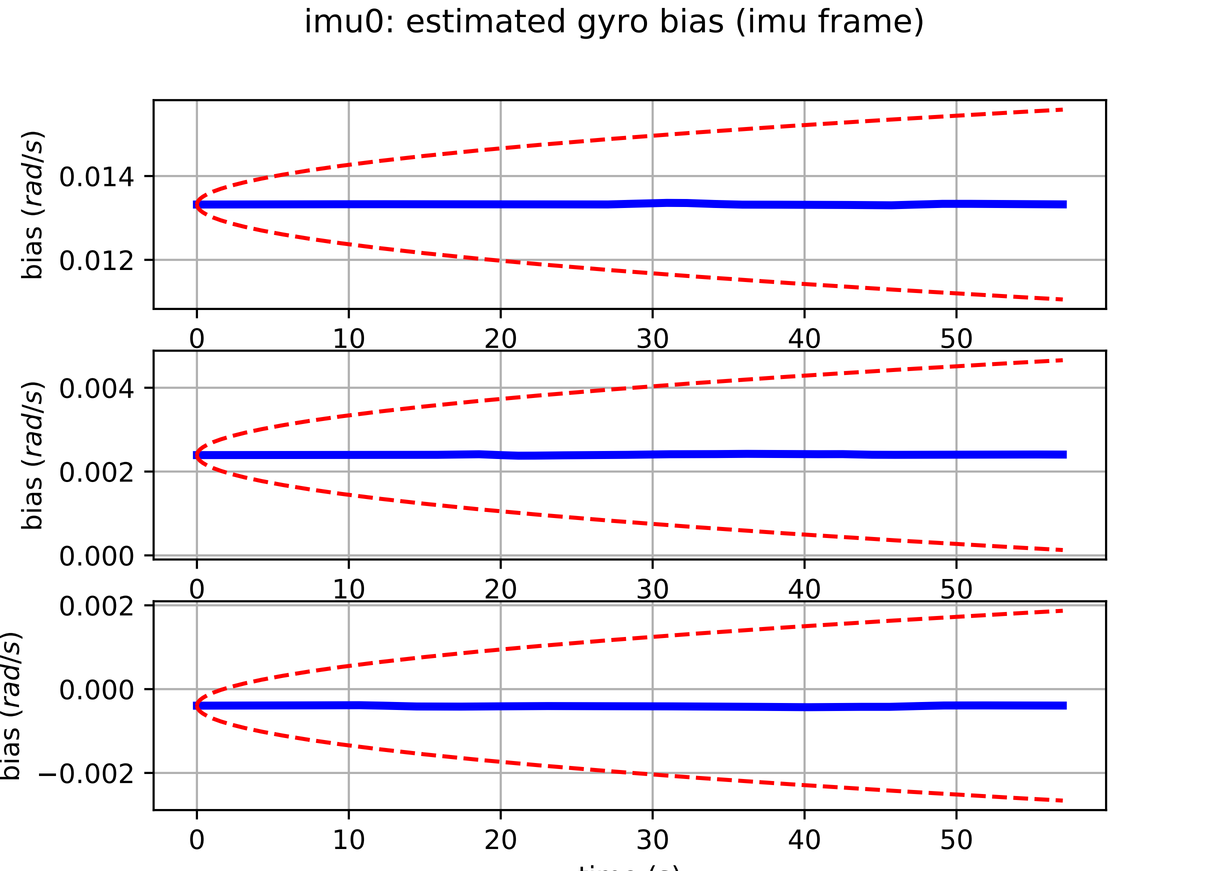


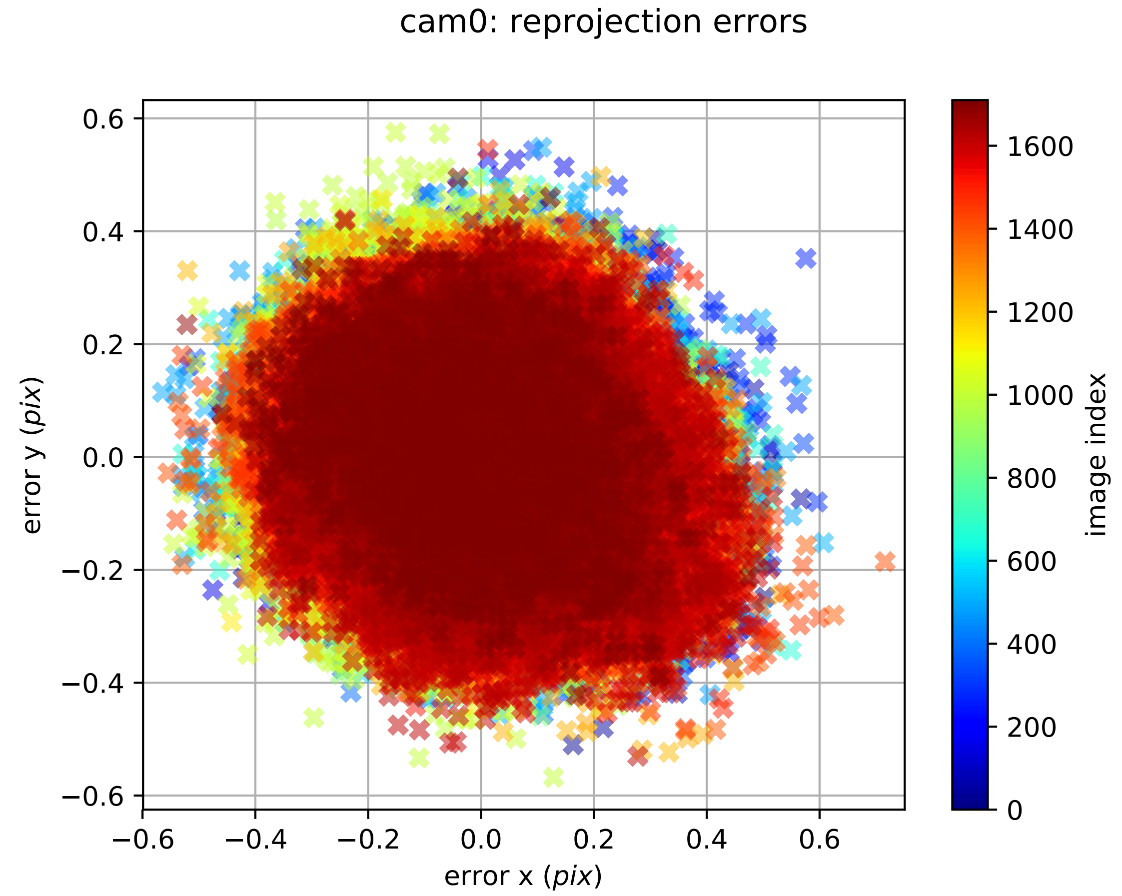
imu0: acceleration error
error (m/s^2)
error (m/s^2)
error (m/s^2)
1
0
−1
0
10000
20000
30000
40000
50000
error index


imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
0.02
0.00
−0.02
bias (m/s^2)
0.00
−0.02
bias (m/s^2)
0.20
0.18
0.16
0
10
20
30
40
50
time (s)

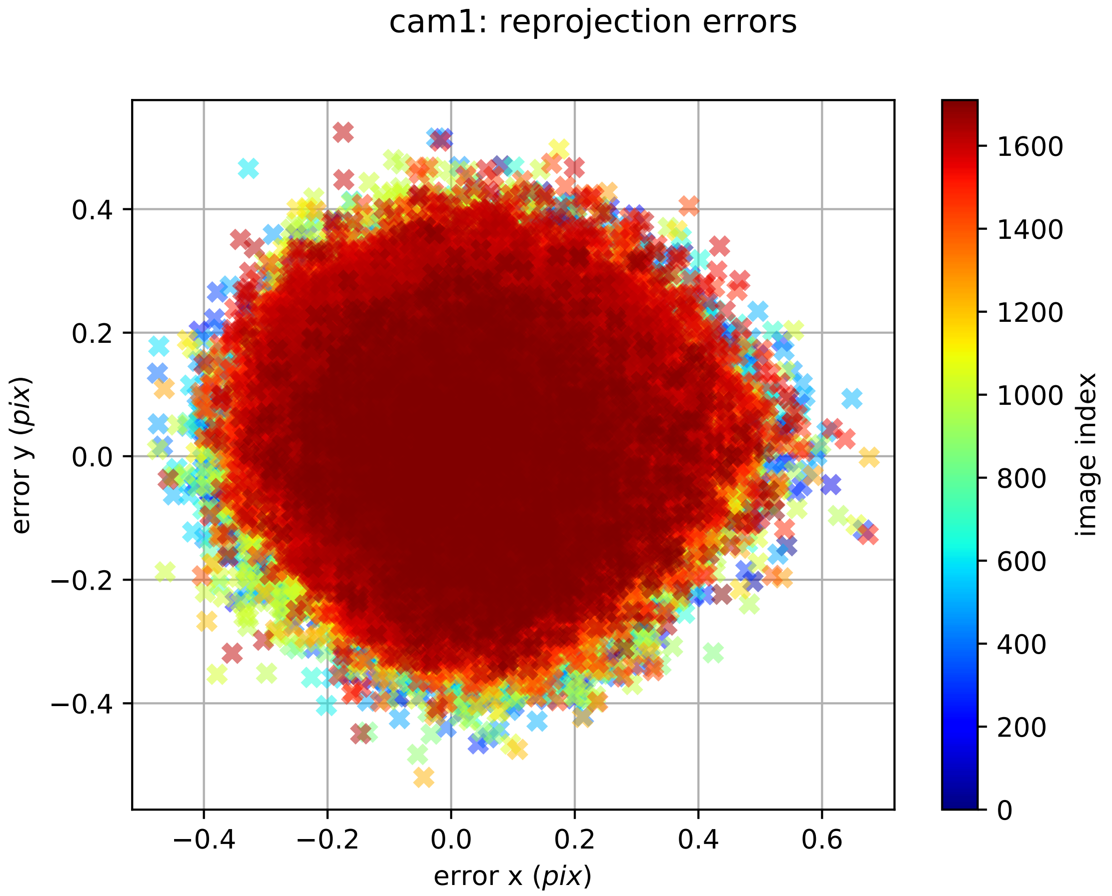

Comparison of predicted and measured angular velocities (body frame)
ang. velocity (rad/s)
est. bodyspline
imu0
0.5
0.0
−0.5
0
10
20
30
40
50
1
0
−1
time (s)


imu0: angular velocities error
error (rad/s)
0.1
0.0
−0.1
0
10000
20000
30000
40000
50000
error index


imu0: estimated gyro bias (imu frame)
bias (rad/s)
0.014
0.012
0.004
0.002
0.000
0.002
0.000
−0.002
0
10
20
30
40
50
time (s)



cam0: reprojection errors
error y (pix)
error x (pix)
image index
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
0.6
1600
1400
1200
1000
800
600
400
200
0


cam1: reprojection errors
error y (pix)
error x (pix)
image index
0.4
0.2
0.0
−0.2
−0.4
0.6
1600
1400
1200
1000
800
600
400
200
0